

Understanding Air Quality Through Data

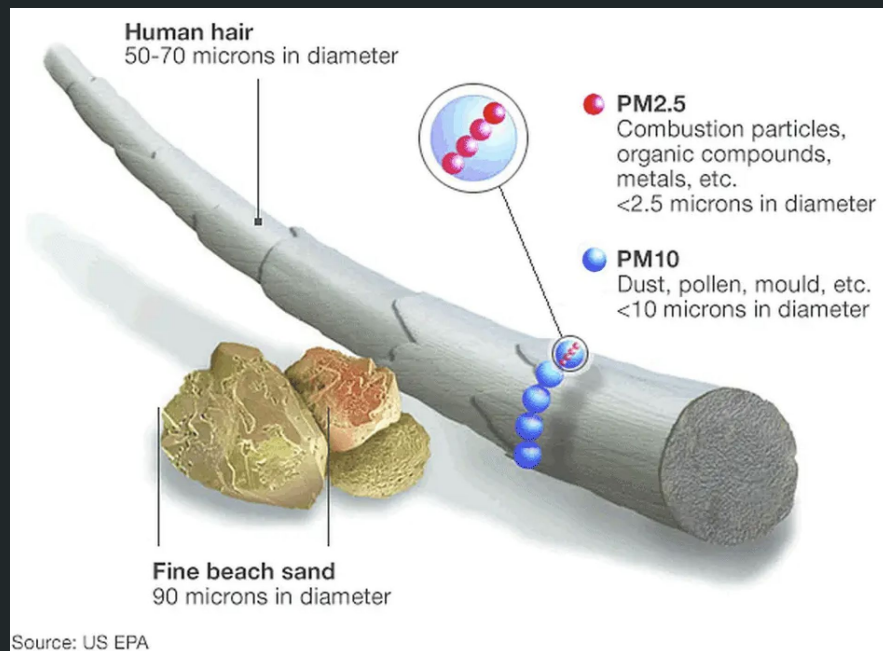
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The CLEAR lab

- CLEAR = Community and Local Engaged Air Research
- Multidisciplinary student/faculty air quality research team
- Goals:
 - air quality (AQ) sensor installation
 - analyzing/mapping sensor data
 - research and public education

PM2.5

- Particulate matter: suspended air particles
- PM2.5: particulate matter < 2.5 μm in diameter
- Smaller particles pose highest health risk
- Sources: fossil fuels, industry, vehicle exhaust, wildfire smoke



PurpleAir Sensors

- PurpleAir: affordable AQ sensor manufacturing company
- Offers smaller, more portable, higher-granularity sensors than EPA
- 92 sensors in Chicago, 62 of which are active



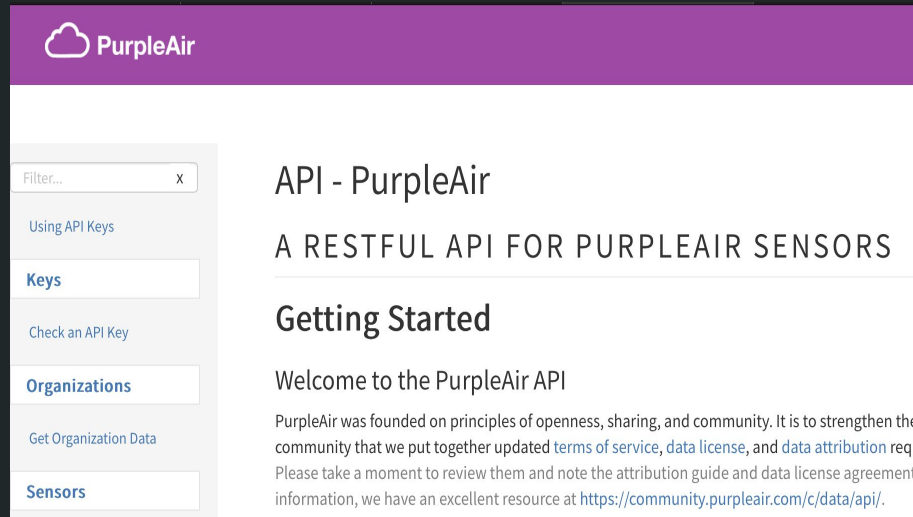
PurpleAir Sensors

- work both indoors and outdoors
- Measure PM2.5, humidity, temperature, and atmospheric pressure
- Readings uploaded to cloud based server, accessible via API



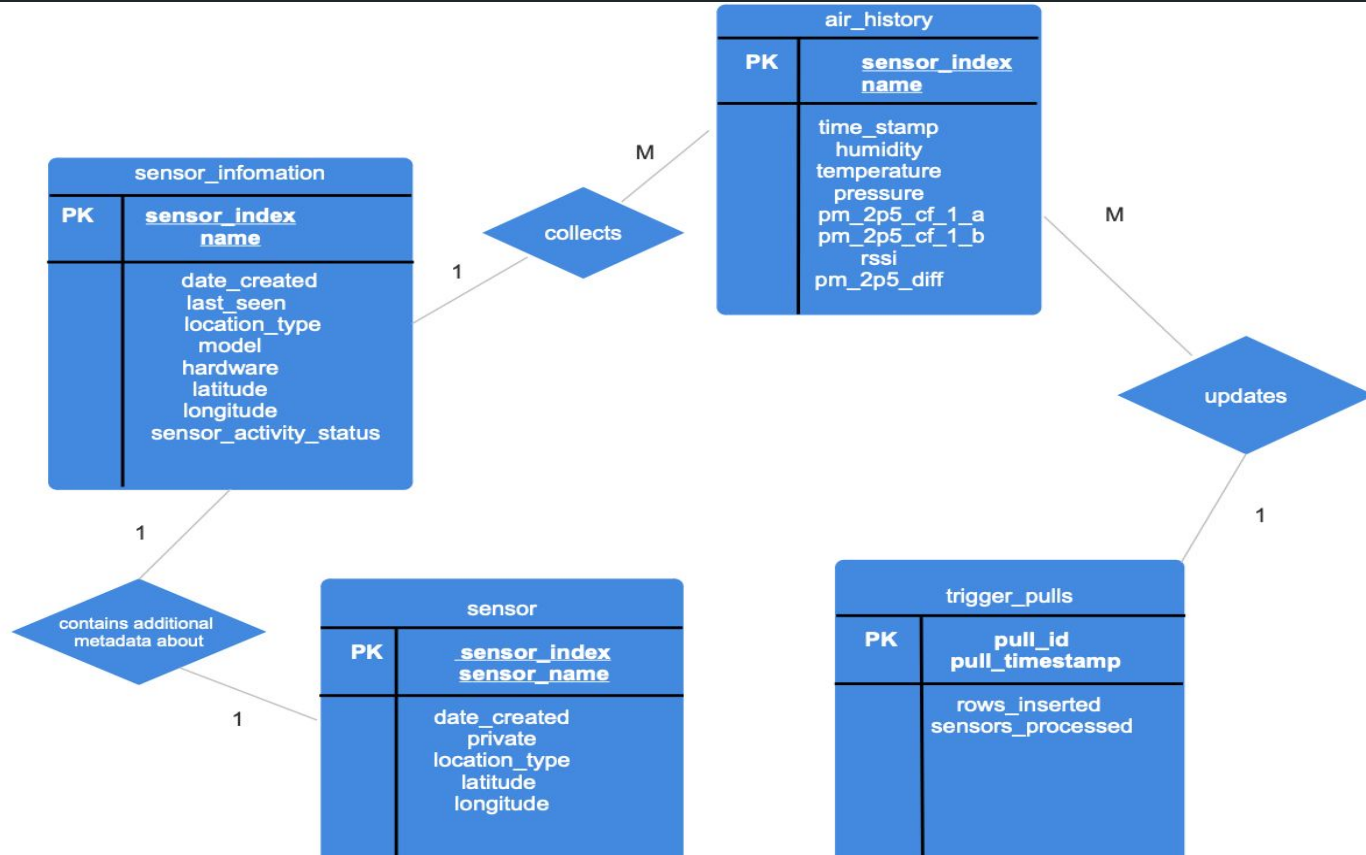
PurpleAir API

- API = Application Programming Interface
- Allows connection between servers/applications

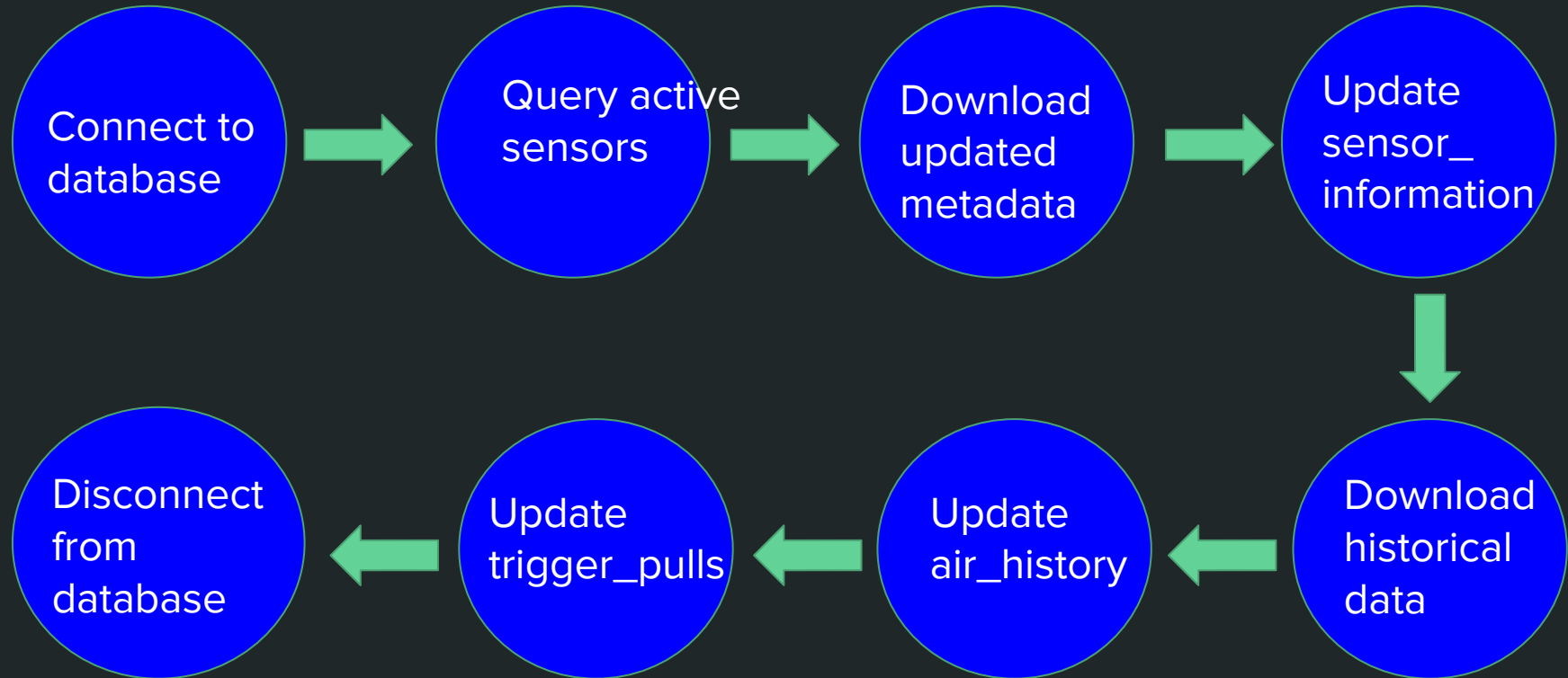


The screenshot shows the PurpleAir API website. The header is purple with the PurpleAir logo (a cloud icon) and the text "PurpleAir". Below the header, there is a sidebar on the left with a search bar labeled "Filter..." and a list of navigation links: "Using API Keys", "Keys" (highlighted in blue), "Check an API Key", "Organizations" (highlighted in blue), "Get Organization Data", and "Sensors" (highlighted in blue). The main content area on the right has the title "API - PurpleAir" and the subtitle "A RESTFUL API FOR PURPLEAIR SENSORS". Below this is a section titled "Getting Started" with the heading "Welcome to the PurpleAir API". The text in this section states: "PurpleAir was founded on principles of openness, sharing, and community. It is to strengthen the community that we put together updated [terms of service](#), [data license](#), and [data attribution](#) requirements. Please take a moment to review them and note the attribution guide and data license agreement. For more information, we have an excellent resource at <https://community.purpleair.com/c/data/api/>."

Database Structure



Database update script



Creating a trigger

- Trigger=event or condition that “triggers” code to automatically run
- Eliminates the need for manual execution

```
> <?xml version="1.0" encoding="UTF-8"?>
> <!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://apple.com">
> <plist version="1.0">
>   <dict>
>     <key>Label</key>
>     <string>com.user.airquality</string>
>     <key>ProgramArguments</key>
>     <array>
>       <string>/Library/Frameworks/R.framework/Resources/bin/Rscript</string>
>       <string>/Users/Shared/USRE_Air_Quality_project/24_hr_update_script.R</string>
>     </array>
>     <key>EnvironmentVariables</key>
>     <dict>
>       <key>PATH</key>
>       <string>/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin:/opt/homebrew/bin</string>
>     </dict>
>     <key>StartCalendarInterval</key>
>     <dict>
>       <key>Hour</key>
>       <integer>0</integer>
>       <key>Minute</key>
>       <integer>0</integer>
>     </dict>
>     <key>StandardOutPath</key>
>     <string>/Users/Shared/USRE_Air_Quality_project/run.log</string>
>     <key>StandardErrorPath</key>
>     <string>/Users/Shared/USRE_Air_Quality_project/run.log</string>
>   </dict>
> </plist>
> EOF
```

Future goals

- Create databases for sensors from other companies (AirGradient, Clarity)
- Continued sensor installation
- Create a public sensor dashboard

What I learned

- Introduction to SQL programming
- Queries and database operations
- ER diagrams for databases

Database update script

